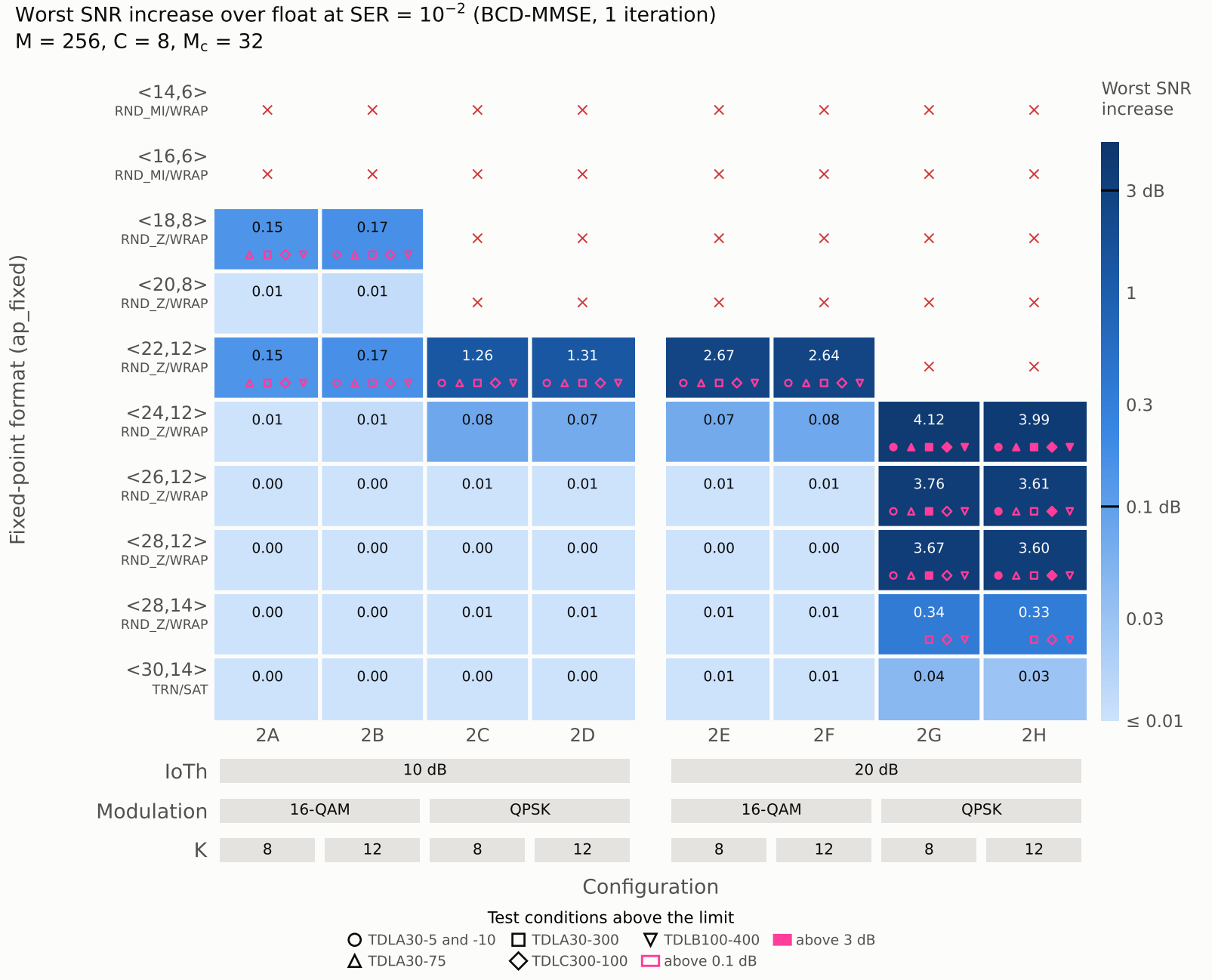
Worst SNR increase over float at SER = 10^{-2} (BCD-MMSE, 1 iteration)
M = 256, C = 8, M_c = 32
Fixed-point format (ap_fixed)
<14,6> RND_MI/WRAP
<16,6> RND_MI/WRAP
<18,8> RND_Z/WRAP
<20,8> RND_Z/WRAP
<22,12> RND_Z/WRAP
<24,12> RND_Z/WRAP
<26,12> RND_Z/WRAP
<28,12> RND_Z/WRAP
<28,14> RND_Z/WRAP
<30,14> TRN/SAT
0.15 0.17 × × × × × ×
× × × × × × × ×
0.15 0.17 × × × × × ×
0.01 0.01 × × × × × ×
0.15 0.17 1.26 1.31 2.67 2.64 × ×
0.01 0.01 0.08 0.07 0.07 0.08 4.12 3.99
0.00 0.00 0.01 0.01 0.01 0.01 3.76 3.61
0.00 0.00 0.00 0.00 0.00 0.00 3.67 3.60
0.00 0.00 0.01 0.01 0.01 0.01 0.34 0.33
0.00 0.00 0.00 0.00 0.01 0.01 0.04 0.03
2A 2B 2C 2D 2E 2F 2G 2H
IoTh 10 dB 20 dB
Modulation 16-QAM QPSK 16-QAM QPSK
K 8 12 8 12 8 12 8 12
Configuration
Worst SNR increase
3 dB
1
0.3
0.1 dB
0.03
≤ 0.01
Test conditions above the limit
TDLA30-5 and -10
TDLA30-300
TDLB100-400
above 3 dB
TDLA30-75
TDLC300-100
above 0.1 dB